

Aptitudinal Alignment Institutional Fidelity Field Theory

Alignment.V1

Aptitude Systems

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Canonical successor to the Beacon Scoring Engine specification.

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Introduction

Aptitudinal Alignment measures how closely a public institution’s documented conduct aligns with what that institution is capable of and obligated to achieve. It is designed as a reverse-FICO system for public entities: instead of scoring an individual’s credit risk, every eligible *entity* — a judge, court, agency, office, or institution — receives one explainable *Alignment Score* derived from the public record.

Each public record, decision, or procedural act is treated as a discrete unit of institutional behavior. These units interact through measurable fields of fidelity, consistency, transparency, efficiency, equity, and accountability. The emergent field defines the Alignment Score.

Product Language

- The page and system name is **Aptitudinal Alignment**.
- The single entity result is the **Alignment Score**.
- The six explainable components are **dimension scores**.
- The interpreted category is the **Alignment Status**.
- Each scored subject is an **entity** with a versioned **potential profile**.

The term “Composite Score” is not a product term. Wherever source material said “Aptitude Score” or “composite,” the unified manual says “Alignment Score.”

Design Principles

- The quantum model is an analytical metaphor and data abstraction — not literal quantum computation.
- Public institutional records are the primary and sufficient evidentiary basis.
- A score is never invented when evidence thresholds are not met; the result is `insufficient_data` with explicit reasons.
- Raw inputs and derived outputs are preserved separately so every score can be recomputed.
- Legal-intelligence conclusions and institutional scores are separate bounded products.
- No Alignment Score is a finding of individual wrongdoing, a prediction of future conduct, or a substitute for an official finding.

Canonical Vocabulary

One vocabulary is used across the codebase, the API, and this manual.

Term	Meaning
Evidence item	A sourced institutional event, record, decision, filing, or report.
Source record	Citation and provenance metadata for an evidence item: source type, credibility tier, date, jurisdiction, and retrieval details.
Behavioral quantum	The normalized analytical representation of an evidence item. It is a data abstraction, not a physical particle.

Term	Meaning
Metric observation	A raw measurement mapped to a dimension and sub-metric.
Dimension score	A reproducible score for Fidelity, Consistency, Transparency, Efficiency, Equity, or Accountability.
Alignment Score	The universal FICO-like result for an entity: a weighted, explainable measure of how closely documented conduct approaches that entity's defined potential after evidence and hard-floor checks.
Entity	Any subject evaluated by the system: judge, court, agency, office, or other institution.
Potential profile	The versioned standard for what an entity can and should achieve, based on mandate, role, jurisdiction, peer class, resources, and public obligations.
Alignment Status	The interpreted category of the Alignment Score: <code>high_alignment</code> , <code>adequate</code> , <code>at_risk</code> , <code>low_alignment</code> , <code>critical_failure</code> , <code>insufficient_data</code> , or <code>suppressed</code> .

Dimension Symbols

The six dimensions use unambiguous symbols in code and documentation: *F* Fidelity, *C* Consistency, *T* Transparency, *E* Efficiency, *Q* Equity, and *Ac* Accountability. *Q* means Equity; *Ac* means Accountability. No other meaning is attached to these symbols.

Evidence and Provenance Model

Both the legal-intelligence pipeline and the Alignment scoring pipeline use one shared evidence foundation. Each evidence item identifies the institution or actor, the event, the source record, source type and credibility tier, citation, document date, jurisdiction, dimension and sub-metric mapping, raw and normalized values, uncertainty, and the processing, ontology, metric registry, and normalization versions that produced it.

Raw evidence is immutable. Derived observations and scores are append-only revisions linked to the input hashes and configuration versions used to produce them.

Source Type Allowlist

Only records with these `source_type` values are eligible for institutional score computation:

```
public_record, regulatory_filing, court_decision, agency_report,
cfpb_complaint, ftc_action, inspector_general, individual_submission
```

Records with `source_type = individual_submission` are supplementary (credibility tier 6, weight 0.05) and cannot dominate any dimension unless corroborated by a public record. Their absence never degrades institutional score validity.

Provenance Fields

```
{
  "id": "hash(entityId + eventId + sourceRecordId)",
  "entity_id": "uuid",
  "event_id": "uuid",
```

```

"source_record_id": "uuid",
"source_type": "court_decision",
"credibility_tier": 1,
"citation": "State v. Example, 123 Or App 45 (2024)",
"document_date": "2024-03-01",
"jurisdiction_code": "OR",
"dimension": "F",
"raw_value": 0.91,
"normalized_value": 0.91,
"confidence": 1.0,
"processing_version": "brain-v1",
"ontology_version": "ontology-v1",
"metric_registry_version": "metrics-v1",
"normalization_version": "norm-v1",
"input_hash": "hex",
"recorded_at": "ISO-8601"
}

```

Legal Intelligence Pipeline

The legal-intelligence pipeline transforms lived legal experiences into structured, queryable, auditable legal intelligence. It is a separate bounded product from institutional scoring.

Flow

```

narrative -> structured event -> claims/doctrines
          -> authority graph and retrieval -> procedural pathways

```

Agents

- **Intake Parser:** structures observable facts only; never infers legal conclusions.
- **Legal Classifier:** maps to claims and doctrines using the ontology only; conservative classification.
- **Graph Seeder:** builds the legal knowledge graph with typed edges (supports, governed_by, codified_in, violates, resolved_by).
- **Vector Retrieval:** returns ranked supporting authority with metadata — supporting material only, no conclusions.
- **Procedural Engine:** builds ordered enforcement pathways with prerequisites and jurisdiction alignment.

Invariants

- No output without provenance.
- No citation without source.
- No authority without jurisdiction.
- No overwrite of historical records.
- No legal conclusion without supporting authorities.
- An incident never silently becomes an institutional score.

Aptitudinal Alignment Scoring

Flow

```
entity + potential profile + evidence
-> metric observations
-> normalization
-> six dimension evaluators
-> hard floors
-> Alignment Score
-> tier / status
```

Every supported entity receives the same explainable score contract, but not the same metric registry. A judge, court, agency, prosecutor's office, or credit reporting agency gets a potential profile appropriate to its mandate, role, jurisdiction, peer class, resources, and public obligations.

Universal Entity Contract

```
{
  "entity_id": "uuid",
  "entity_type": "judge | court | agency | office | institution",
  "potential_profile_version": "string",
  "alignment_score": 0.0,
  "alignment_status": "high_alignment | adequate | at_risk | low_alignment
    | critical_failure | insufficient_data | suppressed",
  "dimensions": { "F": 0.0, "C": 0.0, "T": 0.0, "E": 0.0, "Q": 0.0, "Ac":
    0.0 },
  "evidence_summary": { "count": 0, "source_types": [], "
    missing_requirements": [] },
  "flags": [],
  "audit_reference": "string"
}
```

The numeric range is normalized to [0,1] internally. A FICO-like integer scale may be used only through a documented presentation transform; it must never hide the six dimensions, evidence sufficiency, or limitations.

Default Weights

$$A = \sum_{i=1}^6 w_i \cdot D_i \quad \sum_{i=1}^6 w_i = 1.0$$

Dimension	Symbol	Default weight	Rationale
Fidelity	<i>F</i>	0.22	Legal mandate adherence is foundational
Consistency	<i>C</i>	0.18	Variance in outcomes signals systemic bias
Transparency	<i>T</i>	0.17	Public record access is a legal right
Efficiency	<i>E</i>	0.15	Resource stewardship proxies competence
Equity	<i>Q</i>	0.16	Demographic fairness is a constitutional floor
Accountability	<i>Ac</i>	0.12	Self-correction capacity determines trajectory

Classification Tiers

Tier	Score range	Alignment Status
I	0.85 – 1.00	high_alignment
II	0.70 – 0.84	adequate
III	0.50 – 0.69	at_risk
IV	0.30 – 0.49	low_alignment
V	0.00 – 0.29	critical_failure

Evidence Thresholds and Insufficient Data

A score is emitted only when the configured evidence thresholds pass: at least 12 behavioral quanta per dimension from at least 3 distinct source types. When any dimension fails its threshold, the result is `insufficient_data`: `alignment_score` is null, the failing dimensions are null, and `evidence_summary.missing_requirements` explains why. The system never estimates a missing dimension.

Hard Floors

Equity hard floors override the weighted average and cannot be bypassed:

- Finding of intentional discrimination within 10 years $\Rightarrow Q_{\min} = 0.15$
- Active consent decree for a civil rights violation $\Rightarrow Q_{\min} = 0.25$
- Pattern-or-practice finding $\Rightarrow Q_{\min} = 0.20$

All floor applications are recorded in the score record before the Alignment Score is computed.

Normalization

$$D_{\text{norm}} = \frac{D_{\text{raw}} - D_{\min}}{D_{\max} - D_{\min}}$$

Bounds are configurable per entity class. Values are clamped to $[0, 1]$ — no extrapolation permitted.

Reproducibility

A reproducibility record stores input hashes, configuration versions, code version, and output hash. Any score can be re-derived byte-for-byte from the quantum set and the potential profile version.

Six Dimensions and Metric Registry

Each dimension is defined once. The operational sub-metric tables, weights, source lists, and agent specifications come from the Beacon Scoring Engine specification (source: `Bias Beacon Documentation.tex`, the second of the two LaTeX manuals on the maintainer’s desktop), reproduced verbatim where they define behavior. Only the rules that change scoring behavior are restated here; full metric tables live in the source specification and its metric registry.

Fidelity (F)

Measures whether an institution adheres to its legal mandate, enabling statutes, and constitutional obligations. Sub-metrics: Legal Mandate Compliance Rate (LMCR), Policy Adherence Score (PAS), Regulatory Violation Reversal Rate (RVR), and Statutory Override Frequency (SOF, inverted). Consent orders count as negative compliance evidence regardless of whether the institution agreed to them. Self-reported compliance documentation receives a 0.50 credibility discount unless corroborated. A single court finding of *ultra vires* action floors LMCR at 0.20 for the affected period.

Consistency (C)

Measures variance in institutional outcomes across substantively similar cases. Sub-metrics: Outcome Variance Index (OVI, inverted), Precedent Adherence Rate (PAR), Cross-Case Disparity Score (CDS, inverted), and Decision Reversal Rate (DRR, inverted). A demographic disparity above 15% in any protected class triggers an equity cross-flag regardless of the overall C score. Institutions under consent decrees carry permanent low-consistency markers until a 5-year clean record is established.

Transparency (T)

Measures whether the institution makes its records, decisions, and procedures accessible. Sub-metrics: Record Disclosure Rate (RDR), Response Time Index (RTI), FOIA/Public Records Compliance Rate (FCR), and Proactive Access Index (PAX). Response time is measured from institutional *receipt*, not acknowledgment. Institutions requiring in-person filing only receive an automatic PAX deduction of 0.10. An institution held in contempt for disclosure failure within the last 5 years cannot score above $T = 0.40$.

Efficiency (E)

Measures how effectively an institution converts public resources into mandated outcomes. Sub-metrics: Case Resolution Rate (CRR), Processing Time Score (PTS), Resource Utilization Index (RUI, net of litigation costs), and Backlog Clearance Rate (BCR). Administrative closures without documented outcome do not count. Efficiency scores are always displayed with the peer-class benchmark visible. Budget cuts exceeding 20% in a single year trigger a documented 0.10 grace adjustment for that year only.

Equity (Q)

Measures whether the institution delivers its mandate fairly across demographic groups, geographic areas, and economic strata. Sub-metrics: Demographic Disparity Index (DDI, inverted), Geographic Access Score (GAS), Protected Class Outcome Rate (PCOR), and Appeals Success Rate for protected classes (ASR). Hard floors are defined in Section 5. A minimum $n = 100$ per demographic cell is required before statistical claims; the system never imputes demographic data. A Tier V equity score triggers a mandatory regulatory referral workflow regardless of the overall Alignment Score.

Accountability (Ac)

Measures the institution's capacity to identify its own failures, correct them, and institutionalize reforms. Sub-metrics: Self-Correction Rate (SCR, with a $1.5\times$ multiplier for proactive correction), Audit Compliance (AC), Complaint Resolution Rate (CRR_c, distinct from Efficiency CRR), and Remediation Action Rate (RAR). A pattern of recurring identical violations (3+ consecutive periods) caps Ac at 0.35. Complaints marked resolved without documented resolution trigger a RESOLUTION_FRAUD flag and an audit of all complaint-resolution scores for the preceding 24 months.

Source Credibility and Normalization Bounds

Credibility Hierarchy

Conflicting quanta from multiple sources are resolved by tier weight:

Tier	Weight	Source description
1	1.00	Court orders, consent decrees, regulatory enforcement actions
2	0.90	IG reports, GAO findings, examination summaries, sworn testimony
3	0.75	Aggregated complaint databases, formal complaints
4	0.50	Self-reported public filings (cross-validation required)
5	0.20	Third-party research and reporting (corroboration required)
6	0.05	Individual submissions — supplementary only

Normalization Bounds

Normalization bounds are configured per entity class in the metric registry (e.g., the NCRA peer class bounds table from `Bias Beacon Documentation.tex` for credit reporting agencies). Every bound records its calibration rationale. Values are clamped to $[0, 1]$ — no extrapolation permitted.

Governance and Reproducibility

- Public institutional records are the primary and sufficient evidentiary basis; individual submissions are supplementary.
- Every claim and score component carries a citation chain.
- Raw evidence is immutable; derived outputs are append-only revisions.
- Hard floors are applied and logged before the Alignment Score is computed.
- A reproducibility record stores input hashes, configuration versions, code version, and output hash.
- No Alignment Score is a finding of individual wrongdoing, a prediction of future conduct, or a substitute for an official finding.

Vertical Taxonomy

Alignment.V1 uses one universal evidence and scoring framework across multiple verticals. Vertical profiles describe differences in mandate, legal duties, peer groups, data availability, and valid metrics; they do not create separate scoring engines.

Top-Level Families

1. **Legislative:** legislatures, councils, committees, and related public oversight bodies.
2. **Executive:** departments, agencies, regulators, public administrators, health authorities, human services, and corrections.
3. **Judicial:** judges, courts, clerks, prosecutors, and other adjudicative or justice-system entities.

4. **Consumer-facing and commercial:** credit reporting agencies, lenders, insurers, employers, platforms, utilities, and other entities whose conduct directly affects consumers.

These families are an organizational taxonomy, not a presumption that every member has identical metrics. A Department of Corrections and a health authority are both Executive entities but require different potential profiles.

First Validation Verticals

The first cross-vertical validation set is deliberately heterogeneous:

Family	Example profile	Primary validation purpose
Executive	Department of Corrections	custody, safety, due process, rehabilitation, access, and accountability
Consumer-facing	Nationwide Consumer Reporting Agency	data accuracy, dispute handling, transparency, equity, and remediation
Judicial	Oregon circuit court or judge	procedural consistency, timeliness, access, and reasoned decision-making
Executive	Oregon Health Authority	public health administration, access, outcomes, and data stewardship

Equifax is the first planned live entity instance, but the NCRA profile is the reusable vertical artifact. A corrections department is the second validation profile, not an Equifax-specific extension.

Vertical Profile Contract

A vertical profile defines how a class of entities expresses institutional potential. It supplies configuration to the universal engine without changing the evidence, provenance, or score-result contracts.

Profile Shape

```

{
  "profile_id": "ncra-v1",
  "vertical_family": "consumer_facing",
  "entity_class": "nationwide_consumer_reporting_agency",
  "legal_regimes": ["FCRA", "Regulation V"],
  "peer_class": { "id": "ncra-primary", "minimum_size": 3 },
  "dimensions": {
    "F": { "metrics": ["LMCR", "PAS", "RVR", "SOF"] },
    "C": { "metrics": ["OVI", "PAR", "CDS", "DRR"] },
    "T": { "metrics": ["RDR", "RTI", "FCR", "PAX"] },
    "E": { "metrics": ["CRR", "PTS", "RUI", "BCR"] },
    "Q": { "metrics": ["DDI", "GAS", "PCOR", "ASR"] },
    "Ac": { "metrics": ["SCR", "AC", "CRR_c", "RAR"] }
  },
  "normalization_version": "ncra-norm-v1",
  "sufficiency_rules": {},
  "hard_floors": {},
  "caps": {}
}

```

```
}
```

A corrections profile may use different metric codes, sources, peer classes, and thresholds while retaining the same six-dimensional result contract. The engine must reject incomplete profiles rather than silently substituting metrics from another vertical.

Entity and Activity Separation

An entity profile identifies the subject; an activity record identifies what that subject does. This distinction is mandatory for corporate groups and public departments with multiple programs.

```
entity -> legal entity -> activity -> jurisdiction -> legal regime ->
evidence
```

A label such as “debt collector” must therefore be attached only when the source supports the named legal entity, activity, jurisdiction, and applicable period. It must not be generalized across a corporate group without evidence.

Nationwide Consumer Reporting Agency Profile

This section incorporates the detailed Beacon Scoring Engine baseline for the Nationwide Consumer Reporting Agency (NCRA) class. It is a vertical profile, not a special scoring engine for Equifax.

Class Definition

Institution class	Nationwide Consumer Reporting Agency (NCRA)
Regulatory framework	FCRA, 15 U.S.C.
S 1681 et seq.; Regulation V, 12 C.F.R. Part 1022	
Primary regulator	Consumer Financial Protection Bureau
Secondary regulators	FTC, state regulators, and state attorneys general
Primary peer class	Three nationwide CRAs; specialty CRAs are a secondary cohort
Scoring cadence	Quarterly dimensions; annual Alignment Score, subject to evidence sufficiency

Metric Registry

The NCRA profile uses the Beacon metric registry:

Dimension	Metrics
Fidelity	LMCR, PAS, RVR, SOF
Consistency	OVI, PAR, CDS, DRR
Transparency	RDR, RTI, FCR, PAX
Efficiency	CRR, PTS, RUI, BCR
Equity	DDI, GAS, PCOR, ASR
Accountability	SCR, AC, CRR _c , RAR

Each metric retains the Beacon specification’s definition, weight, source requirements, direction, normalization bounds, minimum sample rules, flags, and hard-floor or cap behavior. These must be represented in the executable metric registry before an empirical score is published.

Complete Evidence Inventory

The NCRA intake should be capable of cataloging, without treating every item as score-eligible:

- CFPB enforcement actions, consent orders, complaint data, and public research;
- FTC actions, state attorney general actions, and court decisions;
- SEC filings, annual reports, quarterly filings, legal proceedings, risk disclosures, and cybersecurity disclosures;
- FCRA litigation, class settlements, appellate decisions, and monitor reports;
- dispute timelines, reinvestigation outcomes, furnisher notifications, security freezes, and consumer disclosure records;
- demographic, geographic, language-access, thin-file, and no-file research; and
- remediation, audit, consent-order monitoring, and recidivism records.

Complaint records are observations about reported consumer experiences. They are not, by themselves, adjudicated findings of wrongdoing and cannot be converted directly into a negative Alignment Score component.

Equifax Pilot Scope

The first live instance is Equifax as a corporate group. The profile must separately resolve Equifax Inc., named subsidiaries, business segments, activities, jurisdictions, legal regimes, and effective periods. Debt collection or recovery-management activity is recorded only where a source supports that scoped classification. The pilot must preserve uncertainty rather than assigning the label to every Equifax entity.

The first output is an auditable evidence and readiness profile. A final Alignment Score is emitted only when the NCRA profile's dimensional evidence requirements are satisfied.

Department of Corrections Profile

The Department of Corrections (DOC) profile is the second validation vertical. It demonstrates that the universal engine can evaluate a public custodial institution without importing consumer-credit metrics.

Profile Scope

The DOC profile may cover a state department, facility, division, or program, but the subject and period must be explicit. Facility-level evidence must not be generalized to the entire department without a documented aggregation rule.

Candidate Metric Families

Dimension	Candidate DOC measures
Fidelity	compliance with constitutional, statutory, court, and department mandates
Consistency	comparable disciplinary, classification, grievance, and release decisions
Transparency	public reporting, records access, incident disclosure, and grievance visibility
Efficiency	case processing, staffing capacity, program delivery, and backlog measures
Equity	demographic disparity, disability access, language access, and geographic/facility differences
Accountability	corrective actions, inspections, investigations, remediation, and recurrence

These are candidate families, not approved metrics. Each metric requires a written definition, valid denominator, source inventory, normalization rule, sufficiency threshold, and legal review before scoring.

DOC-Specific Safeguards

- Separate department, facility, contractor, and individual officer entities.
- Distinguish allegations, grievances, investigations, findings, and adjudicated outcomes.
- Never use incarceration or demographic outcome differences without appropriate denominators and documented confounder handling.
- Treat missing or suppressed correctional data as a transparency and evidence-availability question, not automatic proof of misconduct.
- Preserve facility-level and population-level results separately.

The DOC profile is successful only if it runs through the same universal entity, evidence, metric-observation, and score-result contracts as the NCRA profile while producing a distinct, profile-versioned metric registry.

Executable Metric Registry and Cross-Vertical Validation

The detailed Beacon sub-metrics are being converted into versioned executable registries. The registry is the bridge between the normative specification and the scoring engine. It records each metric's dimension, weight, direction, required source types, minimum sample size where applicable, and profile version.

```
metric code -> dimension -> formula -> source requirements
            -> normalization -> sufficiency -> flags/floors
```

The first reusable registries are:

- `ncra-v1`: the 24-metric Nationwide Consumer Reporting Agency registry derived from the Beacon baseline;

- `corrections-v1`: an independent Department of Corrections registry with custody, due process, grievance, access, capacity, and remediation measures.

Both profiles use the same universal entity, evidence, and score-result contracts. Their metric codes, source requirements, peer classes, and interpretive rules differ by design. Equifax is an entity instance of `ncra-v1`; it is not the profile itself.

At this phase, registry validation and profile separation are implemented. Metric-specific formulas, live source adapters, raw snapshot persistence, and empirical calibration remain subsequent implementation phases. No registry fixture alone constitutes a real-world Alignment Score.

Implementation Phases

The following phases are tracked in the repository as they are built. The current implementation status is recorded at the top of each phase.

1. Shared evidence foundation and provenance contracts (built).
2. Universal Alignment Score engine with insufficient-data and hard floors (built).
3. Judiciary filter experience with case-type filtering and the scale-and-rotate panel (built).
4. Identity system: gold A mark, hero first letter, favicon (built).
5. Universal metric registry and vertical profile contract (specified; next implementation slice).
6. NCRA profile and Equifax role-aware intake (specified; first live instance).
7. Department of Corrections profile and cross-vertical fixtures (specified; reproducibility validation).
8. Source adapters, raw snapshots, and catalog discovery (specified).
9. Normalization service with per-class bounds (specified).
10. Public score API (specified).
11. Reproducibility audit tooling (specified).